

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Aayam Varshney

Enrollment No. NJG241475

Jaypee Institute of Information Technology, Noida

Test-I Examination, 2024

B. Tech, I Semester

Course Title: Mathematics-I

Maximum Time: 1 Hr

Course Code: 15B11MA111

Maximum Marks: 20

CO1	Define the basics of matrices and calculus of functions of one or more variables.
CO2	Explain the concepts of calculus, matrices and Laplace transforms.
CO3	Make use of the concepts of matrices, calculus, differential equations and Laplace transform in solving engineering problems.
CO4	Simplify and solve various problems of vector calculus, differential equations and Laplace transforms in engineering problems.

- Note: 1. Attempt all questions.
2. Calculator is not allowed.

Q1. Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 2 & 0 \\ 3 & 1 & 6 & 5 \\ 2 & 2 & 4 & 2 \end{bmatrix}$ by reducing it to echelon form.

[CO1 (Remembering), 2 Marks]

Q2. Show that $\lim_{(x,y) \rightarrow (0,0)} \sin \left(\frac{\sqrt{x^4+y^2}}{2y} \right)$ does not exist. [CO1 (Remembering), 2 Marks]

Q3. Interpret the values of λ and μ so that the following system of equations has unique solution
 $2x + 3y + 5z = 9$; $7x + 3y - 2z = 8$; $2x + 3y + \lambda z = \mu$.

[CO2 (Understanding), 2 Marks]

Q4. Classify the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

[CO2 (Understanding), 2 Marks]

Q5. Outline the symmetric matrix A from the quadratic form

$$Q = 3x_1^2 + 2x_1x_2 - 4x_1x_3 + 4x_2^2 + 2x_3^2 \quad [\text{CO2 (Understanding), 2 Marks}]$$

Q6. If $f(x, y) = \begin{cases} \frac{\sin(x^3 \cdot y^3)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$, then show that $f_x(0, 0) \neq f_y(0, 0)$.

[CO2 (Understanding), 2 Marks]

Q7. Apply Gauss elimination method to solve the following system of equations

$$x - y + 2z = 3; \quad 2x - 3y + z = -2; \quad 2x + y - 3z = -2.$$

[CO3 (Applying), 4 Marks]

Q8. Develop the second order Taylor's series approximation of the function $f(x, y) = \sin(xy)$ about the point $\left(1, \frac{\pi}{2}\right)$. Also identify maximum absolute error in the region $|x - 1| < 0.1$, $|y - \frac{\pi}{2}| < 0.1$.

[CO3 (Applying), 4 Marks]

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Ashish Varshney

Enrollment No 992401030154

Jaypee Institute of Information Technology, Noida

Test-2 Examination, Odd 2024

B.Tech., I Semester

Course Title : Mathematics-1

Maximum Time : 1Hr

Course Code : 15B11MA111

Maximum Marks :20

CO1	define the basics of matrices and calculus of functions of one or more variables.
CO2	explain the concepts of calculus, matrices and Laplace transforms.
CO3	make use of the concepts of matrices, calculus, differential equations and Laplace transform in solving engineering problems.
CO4	simplify and solve various problems of vector calculus, differential equations and Laplace transform in engineering problems.

Note: Attempt all questions in sequence.

Q1. Apply Dirichlet's integral to find the volume of the region bounded by planes $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} \leq 1, x \geq 0, y \geq 0$ and $z \geq 0$. [CO1 (Remembering), 2Marks]

Q2. Interpret whether the functions $u = x^3 - y^3 z^3, v = x^2 + y^2 z^2 + xyz$ and $w = x - yz$ are dependent or not. If dependent, then find the relation among them. [CO2 (Understanding), 2Marks]

Q3. Classify the critical points of the function $f(x, y) = x^2 + y^2 + \frac{2}{x} + \frac{2}{y}$. Also find the maximum or minimum value of the function. [CO2 (Understanding), 3Marks]

Q4. Solve $\int_0^1 \int_y^{2-y} xy \, dx \, dy$ by changing the order of integration. [CO3 (Applying), 3Marks]

Q5. If a fluid flow is given by $\vec{F} = (x + 2y + 4z) \hat{i} + (2x - 3y - z) \hat{j} + (4x + y + 2z) \hat{k}$ then discover the scalar potential function ϕ such that $\vec{F} = \text{grad } \phi$. [CO4 (Analyzing), 3Marks]

Q6. Discover the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point $(2, 1, 3)$ in the direction of $\vec{u} = \hat{i} - 2\hat{k}$. [CO4 (Analyzing), 3Marks]

Q7. Solve $\int_0^\infty \int_0^\infty e^{-(x+y)} \sin\left(\frac{\pi y}{x+y}\right) \, dx \, dy$ by applying the transformation $u = x + y$ and $v = y$. [CO3 (Applying), 4Marks]

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Aayon Varshney

Enrollment No. 992401030154

Jaypee Institute of Information Technology, Noida

End Semester Examination, Odd Sem 2024

B.Tech., I Semester

Course Title: Mathematics-I

Maximum Time: 2 hrs

Course Code: 15B11MA111

Maximum Marks: 35

CO1	define the basics of matrices and calculus of functions of one variable.
CO2	explain the concepts of calculus, matrices and Laplace Transforms.
CO3	make use of the concepts of matrices, calculus, differential equations and Laplace transform in solving engineering problems
CO4	simplify and solve various problem of vector calculus, differential equations and Laplace transform in engineering problems

Note: Attempt all questions. Non-programmable calculators are allowed.

Q1. Outline the complementary function of the following differential equation $y''' - 6y'' + 11y' - 6y = 0$.
[CO2 (Understanding), 2Marks]

Q2. Extend the following function in terms of unit step functions

$f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3. \end{cases}$ Hence, find the Laplace transform of $f(t)$. [CO2 (Understanding), 2Marks]

Q3. Find the inverse Laplace transform of $\frac{3s+7}{s^2-2s-3}$. [CO1 (Remembering), 2Marks]

Q4. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$, then find $\frac{\partial(u,v)}{\partial(x,y)}$. Are u and v functionally related? If so, find the relationship. [CO2 (Understanding), 3Marks]

Q5. Find the value of $\iint r^3 dr d\theta$ over the area bounded between the circles $r = 2 \cos \theta$ and $r = 4 \cos \theta$. [CO1 (Remembering), 3Marks]

Q6. Apply Green's Theorem to evaluate $\oint_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$, where C is the boundary of the region enclosed by x-axis and the upper half of the circle $x^2 + y^2 = 9$. [CO3 (Applying), 3Marks]

Q7. Identify all the eigen values and corresponding linearly independent eigenvectors of the matrix

$A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$. Is the matrix A diagonalizable? Justify your answer. [CO3 (Applying), 4Marks]

Q8. List the extreme value of the following function using Lagrange method of multipliers, $f(x, y, z) = 2x + 3y + z$, subject to $x^2 + y^2 = 5$ and $x + z = 1$. [CO4 (Analyzing), 4Marks]

Q9. Solve the following differential equation $y'' - 2y' + y = xe^x \sin x + 3^x$. [CO3 (Applying), 4Marks]

Q10. Examine the Stokes theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = -y^2 \hat{i} + x \hat{j} + z^2 \hat{k}$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. [CO4 (Analyzing), 4Marks]

Q11. Examine the solution of the following differential equation, using Laplace transform $y'' + 9y = \cos 2t$; $y(0) = 1, y(\frac{\pi}{2}) = -1$. [CO4 (Analyzing), 4Marks]
